

R102 Friction Stir Welding Robot

Operator, Service and Maintenance Manual

1. Introduction

The R102 Friction Stir Welding Robot is an industrial robotic system designed for automated friction stir welding of aircraft wing panels, spars and structural assemblies. The system delivers repeatable weld quality while continuously monitoring process parameters and equipment health.

Engineering Note: Record spindle operating hours, tool replacement history, cooling system inspections, vibration observations and maintenance activities with timestamps and technician identification.

Inspection Criteria: Verify spindle condition, tool integrity, coolant circulation, electrical connections, communication status and compare measured process values against commissioning baselines.

2. System Overview

The welding cell consists of a six-axis robotic manipulator, friction stir welding spindle, servo drives, spindle motor, cooling system, industrial controller, safety PLC, welding fixture, industrial Ethernet interface and telemetry gateway for process monitoring.

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3. Installation and Commissioning

Verify machine anchoring, spindle alignment, cooling circuit integrity, electrical grounding, emergency stop circuits and communication with the Manufacturing Execution System. Perform spindle runout verification and trial welds before production.

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4. Welding Process

The robot receives a production job, positions the welding head, starts spindle rotation, applies programmed axial force and traverses along the weld path while maintaining constant rotational speed and travel speed. After welding, the spindle retracts and the workpiece is released.

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5. Sensor Monitoring

The system continuously monitors welding temperature, spindle torque, coolant flow rate, motor current and spindle vibration. Welding temperature should remain within the qualified process window defined by engineering. A continuous increase in spindle torque together with increasing vibration indicates possible tool wear or bearing degradation. Reduced coolant flow requires immediate inspection of the cooling circuit.

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6. Alarm and Fault Handling

Typical alarms include spindle overload, coolant flow interruption, excessive vibration, welding temperature deviation, communication timeout, servo overload and emergency stop activation. Repeated alarms require engineering investigation before production resumes.

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7. Common Failure Modes

Typical failure modes include spindle bearing wear, tool wear, cooling pump malfunction, spindle motor overheating, coolant blockage, servo drive faults and fixture misalignment. Progressive increases in vibration and spindle torque generally indicate mechanical degradation.

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8. Preventive Maintenance

Daily maintenance includes inspection of the spindle, cooling hoses, tool holder and alarm history. Weekly maintenance includes lubrication, coolant inspection and spindle vibration checks. Monthly activities include spindle alignment verification, motor diagnostics, cooling pump inspection and controller diagnostics. Quarterly maintenance requires detailed inspection of bearings and replacement of worn consumables where necessary.

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9. Troubleshooting

If weld quality deteriorates, inspect spindle torque trends, verify tool condition and confirm adequate coolant flow. If vibration increases, inspect spindle bearings and tool holder alignment. If welding temperature deviates significantly from the qualified range, verify process parameters and cooling performance.

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10. Spare Parts

Recommended spare parts include spindle bearings, welding tools, spindle seals, coolant pump assemblies, coolant filters, servo motors, encoder modules, industrial Ethernet cables and controller backup batteries.

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11. Diagnostic Guidelines

Evaluate long-term trends in welding temperature, spindle torque, vibration, motor current and coolant flow. Multiple parameters drifting simultaneously generally indicate progressive equipment degradation. Compare current observations with historical maintenance records before planning corrective maintenance.

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12. Operational Best Practices

Always verify fixture alignment before production. Replace worn welding tools at scheduled intervals. Maintain clean coolant lines and filters. Review alarm history at the end of every production shift. Record all maintenance and inspection activities for traceability.

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